

Machine Learning - Complete Notes

Introduction

Machine Learning (ML) is a branch of Artificial Intelligence that allows systems to learn from data and improve automatically without being explicitly programmed.

Types of Machine Learning

- 1 Supervised Learning: Uses labeled data (e.g., classification, regression).
- 2 Unsupervised Learning: Finds hidden patterns (e.g., clustering).
- 3 Reinforcement Learning: Learns using rewards and penalties.

Common Algorithms

- 1 Linear Regression
- 2 Logistic Regression
- 3 Decision Trees
- 4 Random Forest
- 5 Support Vector Machine (SVM)
- 6 K-Means Clustering
- 7 Neural Networks / Deep Learning

Important Concepts

- 1 Training vs Testing Data
- 2 Overfitting and Underfitting
- 3 Bias vs Variance
- 4 Accuracy, Precision, Recall, F1 Score
- 5 Feature Engineering

Applications of Machine Learning

- 1 Recommendation Systems (Netflix, Amazon)
- 2 Image Recognition
- 3 Speech Recognition
- 4 Fraud Detection
- 5 Self-driving Cars
- 6 Healthcare Diagnosis

ML Workflow

- 1 Data Collection
- 2 Data Cleaning
- 3 Feature Selection

- 4 Model Training
- 5 Model Evaluation
- 6 Deployment

Advantages

- 1 Automates decision making
- 2 Handles large data efficiently
- 3 Improves with experience

Disadvantages

- 1 Requires large datasets
- 2 Can be biased
- 3 High computational cost